

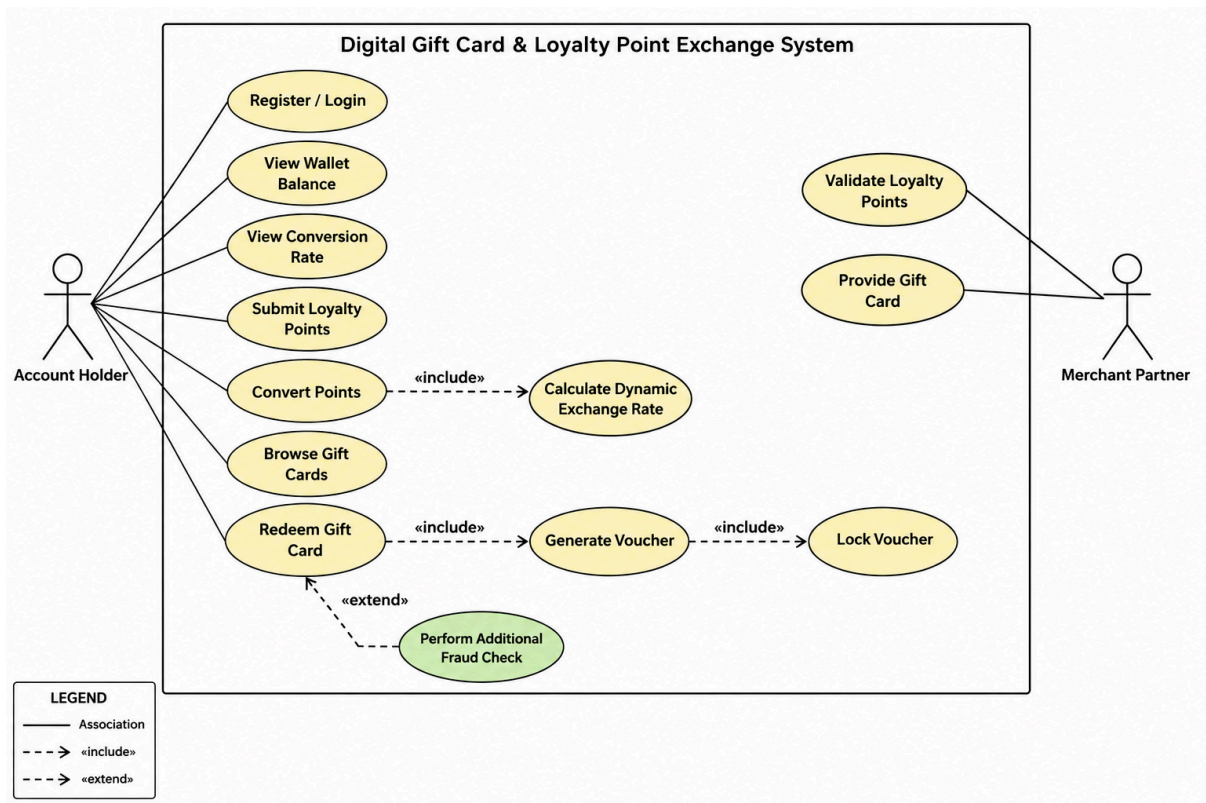
Software Engineering Lab-1

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Section: B

Requirements Table:

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall validate loyalty points submitted by an Account Holder against the corresponding Merchant Partner's records before allowing conversion.	High	Pass: Valid and available loyalty points are accepted for conversion. Fail: Invalid, expired, or insufficient points are rejected.	Validation prevents unauthorized or invalid points from entering the exchange system.
FR-002	Functional	The system shall convert validated loyalty points into universal exchange credits using the currently applicable dynamic conversion rate.	High	Pass: Valid points are converted using the current conversion rate and the resulting credit balance is updated. Fail: Conversion is rejected when the points are invalid or the conversion rate is unavailable.	Users need a common exchange unit to redeem gift cards across participating merchants.
FR-003	Functional	The system shall display the applicable point-to-credit conversion rate and the Account Holder's available universal exchange credit balance before redemption.	Medium	Pass: The current conversion rate and available balance are displayed correctly. Fail: Redemption information is not displayed when the required wallet data is unavailable.	Account Holders need accurate conversion and balance information before making redemption decisions.
FR-004	Functional	The system shall allow an Account Holder to redeem available universal exchange credits for an eligible digital gift card.	High	Pass: A valid redemption request with sufficient credits creates a gift-card voucher. Fail: Redemption is rejected when the Account Holder has insufficient credits or the requested gift card is unavailable.	Gift-card redemption is a core purpose of the digital wallet platform.
FR-005	Functional	The system shall generate a unique single-use digital voucher code and lock the voucher against duplicate redemption.	High	Pass: A unique voucher with a cryptographic checksum is generated and can be redeemed only once. Fail: An expired or previously redeemed voucher is rejected.	Single-use voucher locking prevents duplicate redemption and reduces fraud.
NFR-001	Performance & Security	Point conversion and gift card redemption operations shall complete within 250 ms under the defined normal operating load while maintaining zero ledger balance drift.	High	Pass: Benchmarking tests demonstrate that the operations complete within 250 ms and wallet balances remain consistent under simulated peak load. Fail: The latency target is exceeded or any ledger balance inconsistency occurs.	Fast processing and accurate financial records are essential for a trustworthy exchange platform.
NFR-002	Security	The system shall protect digital voucher codes against unauthorized reuse, tampering, and fraudulent redemption.	High	Pass: Tampered, expired, or previously redeemed vouchers are rejected and valid vouchers can be redeemed only once. Fail: A modified or previously redeemed voucher is accepted.	Strong voucher protection is necessary to prevent fraud and financial loss.

UML Use-Case-Diagram:



Use-Case-Flow:

Use-Case Flow Specification

Use Case: Redeem Digital Gift Card

Use Case ID: UC-01 Primary Actor: Account Holder Supporting Actor: Merchant Partner

Preconditions

1. The Account Holder is authenticated.
2. The Account Holder has a valid digital wallet.
3. The Account Holder has sufficient universal exchange credits.
4. The requested gift card is available.
5. The system is connected to the required merchant/gift-card service.

Postconditions

1. The required universal exchange credits are deducted from the Account Holder's wallet.
2. A unique digital voucher is generated.
3. The voucher is locked for single use.
4. The redemption transaction is recorded in the wallet ledger.
5. The Account Holder receives the digital gift card and voucher.

Main Success Scenario

1. The Account Holder selects the desired digital gift card.
2. The system displays the gift card value and required universal exchange credits.

3. The Account Holder confirms the redemption request.
4. The system verifies that the Account Holder has sufficient exchange credits.
5. The system verifies that the selected gift card is available.
6. The system calculates the applicable redemption amount.
7. The system deducts the required exchange credits from the Account Holder's wallet.
8. The system generates a unique single-use voucher code.
9. The system applies the voucher's security/checksum information.
10. The system locks the voucher against duplicate redemption.
11. The system records the successful transaction in the ledger.
12. The system displays the digital gift card and voucher to the Account Holder.

Alternate Flow — Insufficient Credits

A1. At Step 4, the system determines that the Account Holder does not have sufficient universal exchange credits.

A2. The system rejects the redemption request.

A3. The system does not deduct any credits.

A4. The system informs the Account Holder that the wallet balance is insufficient.

A5. The use case ends without generating a voucher.

Github Repo Link: <https://github.com/AtharvMitt/digital-gift-card-loyalty-exchange>